Lecture 4, Mar 23 2026.

11.1. Sequences

Def. Let consider $f: \mathbb{N} \longrightarrow \mathbb{R}$ then $f(0)=a_0$, $f(1)=a_1$, --, $f(n)=a_n$... i.e, the graph of f is called a sequence.

Notation: $\{a_0, a_1, a_2, -, a_n\}$ is a sequence which is also denoted by $\{a_n\}$ or $\{a_n\}_{n=1}^{\infty}$.

Example. Sometimes de sequences can be defined by giving a formula for the nth term.

- $\left\{\frac{1}{2^n}\right\}_{n=1}^{\infty} = \left\{\frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, --, \frac{1}{2^n}, ---\right\}$.

$$a_1 = \frac{1}{2}, \quad a_2 = \frac{1}{2^2}, \quad --, \quad a_n = \frac{1}{2^n}.$$

- $\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}$ the formula is $a_n = \frac{n}{n+1}$,

$$\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty} = \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, ---\right\}.$$

- $\{\sqrt{3}, \sqrt{4}, \sqrt{5}, \sqrt{6}, --\}$ $\quad a_n = \sqrt{2+n}$.

$$a_1 = \sqrt{3}, \quad a_2 = \sqrt{4}, \quad a_3 = \sqrt{5},$$

- $\left\{(-1)^n \frac{(n+1)}{3^n}\right\}_{n=1}^{\infty} = \left\{-\frac{2}{3}, \frac{3}{3^2}, -\frac{4}{3^3}, ---\right\}$.

Example. Find a formula for the general term a_n of the sequence

$$\left\{\frac{3}{5}, -\frac{4}{25}, \frac{5}{125}, -\frac{6}{625}, \frac{7}{3125}, --\right\}.$$

Sln. We claim that

$$a_n = \frac{(-1)^{n+1}(n+2)}{5^n}$$

The Limit of a Sequence.

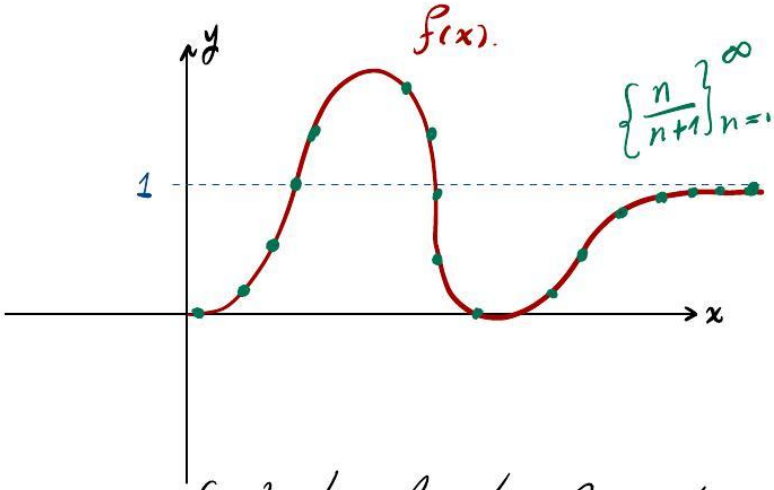
Let consider the sequence

$$\left\{\frac{n}{n+1}\right\} = \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots\right\}.$$

Notice that $f(x) = \dfrac{x}{x+1}$ $\quad Dom\, f = \mathbb{R} - \{-1\}$ then

$$\lim_{x\to\infty} \frac{x}{x+1} \overset{L'H}{=} \lim_{x\to\infty} (1) = 1.$$

So,



then $\left\{\dfrac{n}{n+1}\right\}$ is convergent to 1.

Def. A sequence $\{a_n\}$ has limit L and we write

$$\lim_{n\to\infty} a_n = L \quad \text{or} \quad a_n \to L \text{ as } n \to \infty.$$

if we make the terms a_n as close to L as we like by taking n sufficiently large.

Properties of convergent sequences.

Thm If $\lim_{x\to\infty} f(x) = L$ and $f(n) = a_n$ when $n \in \mathbb{N}$, then

$$\lim_{n\to\infty} a_n = L.$$

Limits law for sequences. if $\{a_n\}$ and $\{b_n\}$ are convergent sequences.

0). $\lim_{n\to\infty} \dfrac{1}{n^r} = 0$ if $r > 0$.

1). $\lim_{n\to\infty} (a_n \pm b_n) = \lim_{n\to\infty} a_n \pm \lim_{n\to\infty} b_n$

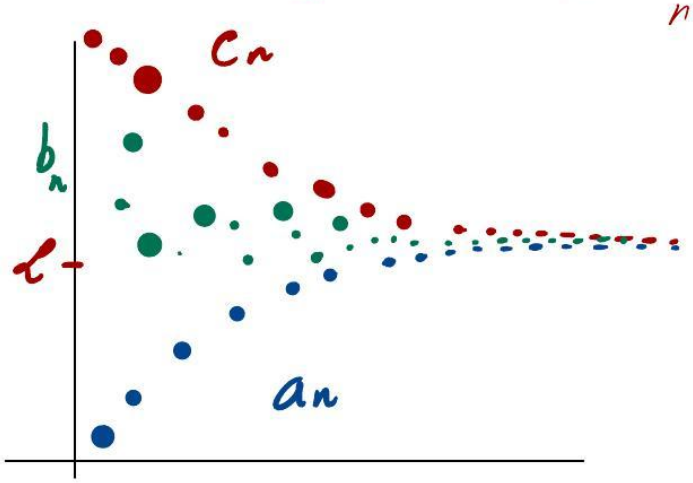
2). $\lim_{n\to\infty}(c\cdot a_n) = c\cdot \lim_{n\to\infty} a_n$.

3). $\lim_{n\to\infty}(a_n\cdot b_n) = \left(\lim_{n\to\infty} a_n\right)\left(\lim_{n\to\infty} b_n\right)$.

4). $\lim_{n\to\infty}\frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}$ if $\lim_{n\to\infty} b_n \neq 0$.

5). $\lim_{n\to\infty} a_n^p = \left[\lim_{n\to\infty} a_n\right]^p$ if $p>0$ and $a_n>0$.

6). (Squeeze thm) if $a_n \le b_n \le c_n$ for $n \ge n_0$ and $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$



then $\lim_{n\to\infty} b_n = L$.

7). If $\lim_{n\to\infty} |a_n| = 0$ then $\lim_{n\to\infty} a_n = 0$.

Examples: a). Is the sequence $a_n = \frac{n}{\sqrt{10+n}}$ convergent or divergent?

This is divergent b/c.

$$\lim_{n\to\infty}\frac{n}{\sqrt{10+n}} = \lim_{n\to\infty}\frac{1}{\sqrt{10/n^2 + 1/n}} = \infty.$$

b). Compute $\lim_{n\to\infty}\frac{\ln n}{n}$.

Sln. We can't apply L'Hospital's rule directly b/c it applies not to the sequences but to function of a real variable.

Let $f(x) = \frac{\ln x}{x}$

$$\lim_{x\to\infty} \frac{\ln x}{x} \overset{L'H}{=} \lim_{x\to\infty} \frac{1}{x} = 0$$

So, by thm $\lim_{n\to\infty} \frac{\ln n}{n} = 0$.

Example. Determinate whether the sequence $a_n = (-1)^n$ is convergent or divergent.

Sln.

$$\{-1, 1, -1, 1, -1, 1, \ldots\}.$$

Since the terms oscillate between 1 and -1 infinitely often, a_n does not approach any number.

Example. Evaluate $\lim_{n\to\infty} \frac{(-1)^n}{n}$ if it exists.

Sln. (ST).

$$-\frac{1}{n} \le \frac{(-1)^n}{n} \le \frac{1}{n} \quad \text{for all } n.$$

$-\frac{1}{n} \to 0$ as $n\to\infty$; $\frac{1}{n} \to 0$ as $n\to\infty$

then, by ST $\lim_{n\to\infty} \frac{(-1)^n}{n} = 0$.

(IT II) $\lim_{n\to\infty} |a_n| = \lim_{n\to\infty} |(-1)^n/n| = \lim_{n\to\infty} \frac{1}{n} = 0$. By thm

$$\lim_{n\to\infty} \frac{(-1)^n}{n} = 0$$

Thm. If $\lim_{n\to\infty} a_n = L$ and the function f is continuous at L, then

$$\lim_{n\to\infty} f(a_n) = f(L).$$

Example. Find $\lim_{n\to\infty} \sin\frac{\pi}{n}$

Sln. Sine function is continuous at 0 and $\lim_{n\to\infty} \frac{\pi}{n} = 0$ so

$$\lim_{n\to\infty} \sin\frac{\pi}{n} = \sin(0) = 0.$$

Example. Discuss the convergence of the sequence $a_n = \frac{n!}{n^n}$ where $n! = 1\cdot 2\cdot 3\cdots n$.

Sln.

$$0 \le a_n = \frac{n!}{n^n} = \frac{1\cdot 2\cdot 3\cdot 4\cdots (n-1)\cdot n}{\underbrace{n\cdot n\cdot n\cdots n}_{n-\text{times}}} = \frac{1}{n}\left(\frac{2\cdot 3\cdot 4\cdots n}{n\cdot n\cdot \quad n}\right) \le \frac{1}{n}$$

Since $\lim_{n\to\infty} \frac{1}{n} = 0$ By S-T we have

$$\lim_{n\to\infty} \frac{n!}{n^n}.$$

Problem For what values of r is the sequence $\{r^n\}$ convergent?